

IF3170 Inteligensi Artifisial

Institut Teknologi Bandung

k-Nearest Neighbor (KNN)

Semester I TA 2026/2027



Similarity



We can recognize them as cats, why?

They have similar “features”

This is basically how **“Nearest Neighbors”** works

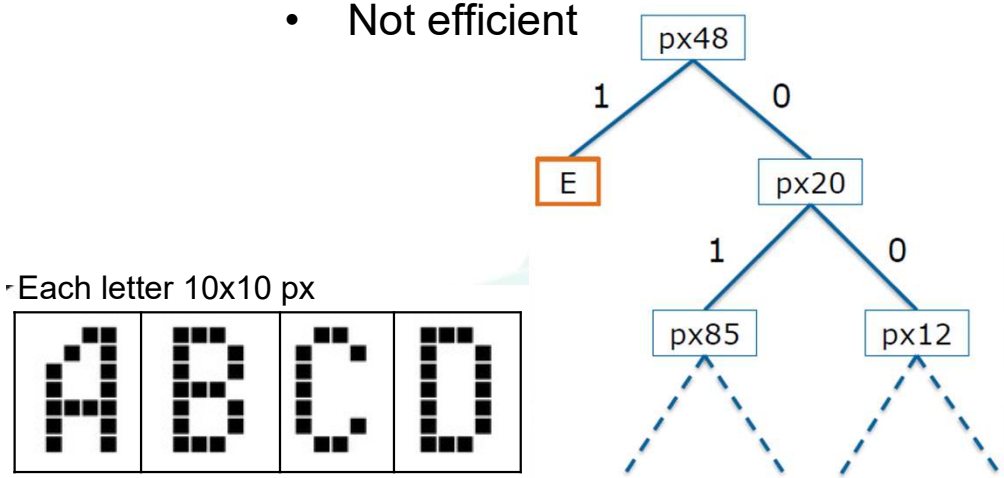
Recall: Decision tree

- Decision tree is good for **limited** and **discrete attributes**
- In case of classification with numerous attributes, the tree resulted might be too complex and prone to overfit

- For example: in image classification

Character recognition (using pixels)
10x10 pixel image

- The tree will be too long
- A complete ID3 will result a 100-level-depth
  tree (100-dimensional input)
- Not efficient



Each letter 10x10 px

k-Nearest Neighbor

- Supervised Learning
- Instance-Based Classifier (Store all training data)
- Lazy learner
- Non parametric

No hypothesis

↓

Unseen data prediction: Find class from similar stored data

Classification (Predict unseen data)

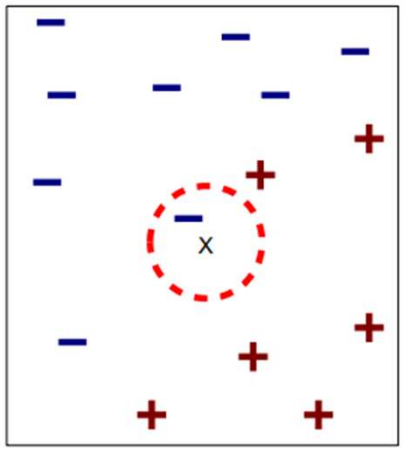
Measures ‘distance’ of query (unseen data) to all instance (in training data)

- Symbolic attribute: 1 (different value), 0 (same value)
- Numeric attribute: Euclidean Distance

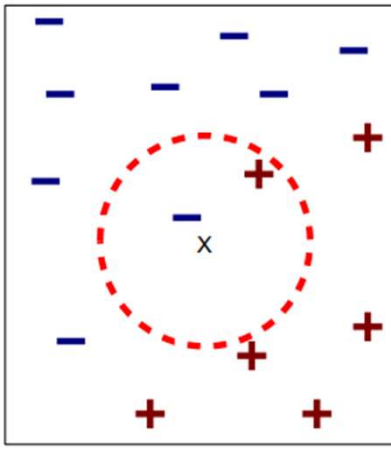
Find k ‘most similar’ instances (k nearest neighbor)

Find the majority class from k nearest neighbor

Class/ Label Prediction: Majority Class of k nearest neighbor



(a) 1-nearest neighbor



(b) 2-nearest neighbor

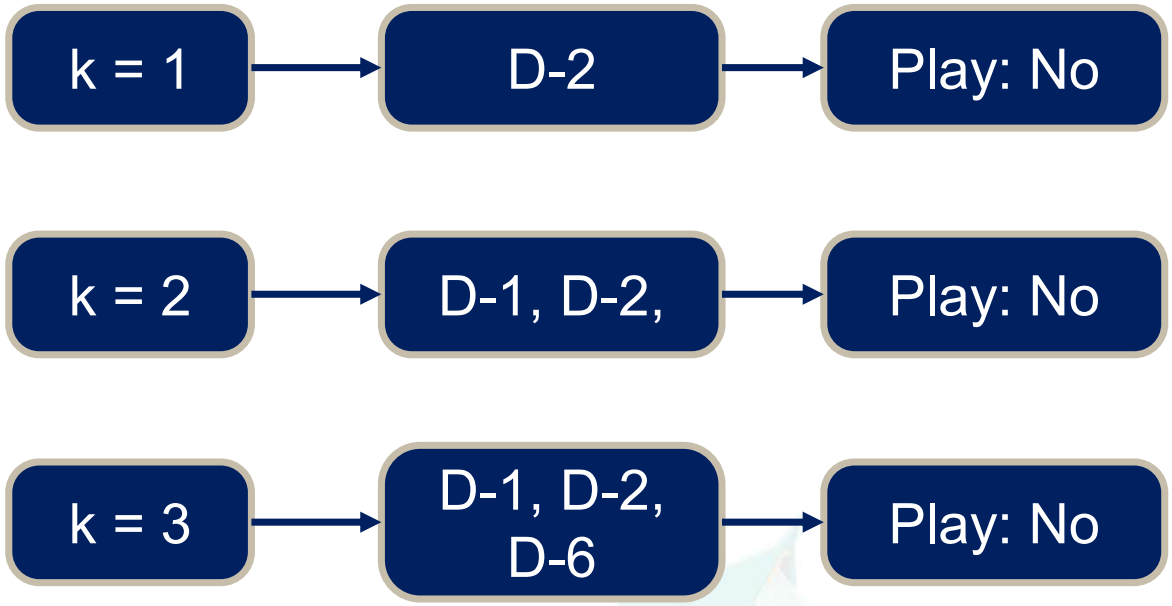
Example: Play Tennis Dataset

outlook	temp.	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes

outlook	temp.	humidity	windy	play
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Classify New Instance: <Sunny, Cool, High, True>

outlook	temp.	humidity	windy	play	Distance
sunny	hot	high	false	no	2
sunny	hot	high	true	no	1
overcast	hot	high	false	yes	3
rainy	mild	high	false	yes	3
rainy	cool	normal	false	yes	3
rainy	cool	normal	true	no	2
overcast	cool	normal	true	yes	2
sunny	mild	high	false	no	2
sunny	cool	normal	false	yes	2
rainy	mild	normal	false	yes	4
sunny	mild	normal	true	yes	2
overcast	mild	high	true	yes	4
overcast	hot	normal	false	yes	2
rainy	mild	high	true	no	



Notes on k-Nearest Neigbbor

- Advantages
  - Approximation can be less complex for complex target function
- Disadvantages
  - Cost of classifying new instance high
  - Consider all features → target function depends only on a few features

K-Nearest Neighbors

Training Process

- Memorize all training data and labels

Testing Process

- Define the class based on similarity of their **K** neighbors
- Similarity is defined by the distance from new data to their **K** neighbors
- Return label of **K** closest one as the new data's label

Distance Calculation

- L1 : Manhattan Distance
- L2 : Euclidean Distance
- etc

id	hobi	umur	pendidikan	kelas
1	Game	remaja	sma	1
2	Game	dewasa	s1	2
3	Game	dewasa	diploma	3
4	Baca	dewasa	s1	3
5	Olahraga	dewasa-muda	s1	2
6	Olahraga	dewasa-muda	diploma	3
7	Game	dewasa-muda	sma	1
8	Olahraga	dewasa-muda	sma	1
9	Baca	dewasa-muda	sma	1
10	Game	dewasa-muda	s1	3
11	Baca	Remaja	diploma	1
12	Game	remaja	diploma	2
13	game	dewasa	sma	3

1. Hitung Jarak setiap instance data latih utk data uji berikut:
   hobi = olahraga; umur = dewasa; pendidikan = s1
2. Untuk k=5, maka kelas dari data uji di atas adalah: …

Id	Jarak thd data baru
1	1+1+1=3
2	**1+0+0=1**
3	**1+0+1=2**
4	**1+0+0=1**
5	**0+1+0=1**
6	**0+1+1=2**
7	1+1+1=3
8	0+1+1=2
9	1+1+1=3
10	1+1+0=2
11	1+1+1=3
12	1+1+1=3
13	1+0+1=2

Distance Measurement on Numeric Attributes

$$D = \left(\sum_{i=1}^{n} |p_i - q_i|^p \right)^{1/p}$$

Minkowski distance

$$D_m = \sum_{i=1}^{n} |p_i - q_i|$$

Manhattan distance

$$D_e = \left(\sum_{i=1}^{n} (p_i - q_i)^2 \right)^{1/2}$$

Euclidean distance

Brightness	Saturation	Class
40	20	Red
50	50	Blue
60	90	Blue
10	25	Red
70	70	Blue
60	10	Red
25	80	Blue

Brightness	Saturation	Class
20	35	?

$d1 = \sqrt{(20 - 40)^2 + (35 - 20)^2}$

$= \sqrt{400 + 225}$

$= \sqrt{625}$

$= 25$

$d2 = \sqrt{(20 - 50)^2 + (35 - 50)^2}$

$= \sqrt{900 + 225}$

$= \sqrt{1125}$

$= 33.54$

$$D_e = \left(\sum_{i=1}^{n} (p_i - q_i)^2 \right)^{1/2}$$

Euclidean distance

$$\sqrt{(X_2-X_1)^2+(Y_2-Y_1)^2}$$

- X_2 = New entry's brightness (20).
- X_1= Existing entry's brightness.
- Y_2 = New entry's saturation (35).
- Y_1 = Existing entry's saturation.

Brightness	Saturation	Class
40	20	Red
50	50	Blue
60	90	Blue
10	25	Red
70	70	Blue
60	10	Red
25	80	Blue

Brightness	Saturation	Class
20	35	?

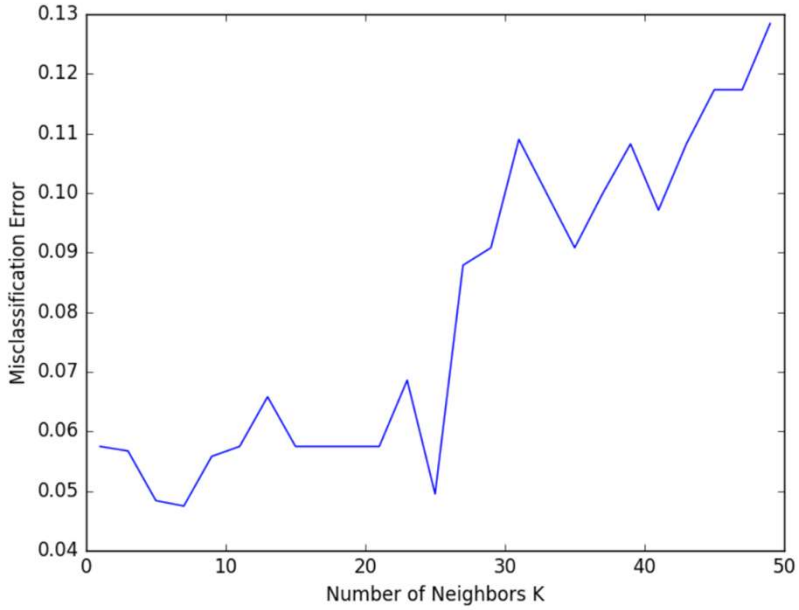
Brightness	Saturation	Class	Distance
40	20	Red	25
50	50	Blue	33.54
60	90	Blue	68.01
10	25	Red	10
70	70	Blue	61.03
60	10	Red	47.17
25	80	Blue	45

Hyperparameters

- What is the best **distance** to use?
- What is the best value of **k** to use?

How?

- Very problem-dependent.
- Must try them all out and see what works best.
- Use train set and validation set to avoid overfitting



Myth: K value should be an odd-number

Instead, test on K-NN Hyperparameter Observation

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THANK YOU